MIL-H-44105
18 July 1983
SUPERSEDING
(See 6.5)

MILITARY SPECIFICATION

HAT, CAMOUFLAGE PATTERN: DESERT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a hat made of cotton/nylon twill material dyed in a 6 color camouflage desert pattern.

1.2 Classification. The hat shall be of one type in the following sizes as specified (see 6.2).

Schedule of sizes

6-3/8	6-7/8	7-3/8
6-1/2	7	7-1/2
6-5/8	7-1/8	7-5/8
6-3/4	7-1/4	7-3/4
		7-7/8

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this Document or by letter.

FSC 8415

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:56Z
Check the source to verify that this is the currentversion before use.

SPECIFICATIONS

FEDERAL

V-L-61	-	Laces, Nylon
KK-L-271	-	Leather, Cattlehide, Strap, Vegetable-Tanned
DDD-L-20	-	Label; For Clothing, Equipage, and Tentage (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-T-43548	-	Thread, Polyester, Cotton-Covered and Rayon-Covered
MIL-C-44034	-	Cloth, Twill, Camouflage Pattern, Cotton and Nylon for Desert Uniform
MIL-C-43836	-	Cloth, Interlining, Nonwoven

STANDARDS

FEDERAL

FED-STD-751	-	Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-1188	-	Commercial Packaging of Supplies and Equipment

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

TEXTILE COLOR CARD ASSOCIATION

Standard Color Card of America

(Application for copies should be addressed to the Textile Color Card Association of the United States, Inc., 24 East 38th Street, New York, NY 10016.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:56Z
Check the source to verify that this is the current version before use.

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Guide sample. Guide samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from this document may appear in the sample, in which case this document shall govern.

3.2 Materials (see 6.4).

3.2.1 Basic material. The basic material shall be a blended cotton/nylon, twill cloth dyed in the US Army 6 color camouflage desert pattern conforming to MIL-C-44034.

3.2.2 Brim binding. The brim binding shall be bias-cut from the basic material (see 3.2.1).

3.2.3 Brim interlining and head-band stay piece. The brim interlining and head-band stay piece shall be a white, nonwoven, continuous filament polyester cloth conforming to type I, class 2 of MIL-C-43836.

3.2.4 Laces, nylon. The nylon lace for the chin strap shall be Tan 337 conforming to type II, class 2, of V-L-61 except that the requirements for length and tipping shall not apply.

3.2.5 Keeper. The chin strap keeper shall be made from vegetable-tanned cattle-hide strap leather, 6/64 to 8/64 inch thick, natural in color, conforming to type I, II, III or IV, class 1 of KK-L-271 except that the requirements for the treatment for fungicide paranitrophenal, crocking, cracking, breaking force, elongation, chloroform extract, water-soluble material, total ash, and degree of tannage shall not apply. A plastic fixlock keeper may be substituted for the leather keeper.

3.2.6 Thread. The thread for seaming and stitching the hat shall be Khaki, P-1, C.A. 66019 polyester/cotton-covered and rayon-covered, ticket Nos. 50, 3 ply, and 70, 2 ply conforming to MIL-T-43548.

3.2.6.1 Colorfastness. The dyed thread shall show fastness to laundering equal to or better than the standard sample. When no standard sample is available, the dyed thread shall show "good" fastness to laundering.

3.2.7 Labels. Each hat shall have a combination identification-size label and an instruction label conforming to DDD-L-20. The label color shall be Khaki, Cable No. 70188 and shall show fastness to laundering in accordance with DDD-L-20 requirements.

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:56Z
Check the source to verify that this is the currentversion before use.

3.2.7.1 Identification-size label. The identification-size label shall conform to type VI, class 4.

3.2.7.2 Instruction label. The instruction label shall conform to type VI, class 3 and include the following information:

CAUTION

Initial loose fit required
for continued comfort.

LAUNDERING

Spot wash badly soiled areas
Wash in hot water (140°F)
using soap or detergent.
Machine spin or squeeze by hand
Machine dry at moderate heat
or hang to dry.

DO NOT STARCH

Suspend headnet from top of brim, drape
headnet over shoulders and attach front
loops to breast pocket buttons.

DO NOT REMOVE THIS LABEL

3.2.8 Ventilators, with washers. The screen type ventilators with washers shall be brass, black chemical, 9/16 inch outside rim diameter, and 7/16 inch outside diameter screening, conforming to Stimpson Company, No. 53 or equal. The contractor shall submit a certificate of compliance for approval (see 4.2.1.1).

3.3 Design. The hat is a light weight soft crowned camouflage model with a standard width quilt stitched brim and chin strap. The hat is lined with self material. (See figure I)

3.4 Patterns. Standard patterns to be used to cut working patterns will be furnished by the Government. The working patterns shall be identical to the standard patterns which shall not be altered in any way. The patterns provide 3/16 inch for seaming side lining to crown, 3/8 inch for seaming brim to sides, and 1/4 inch for seaming all others.

3.4.1 Pattern parts. The components of the hat shall be cut from the materials specified according to the pattern parts indicated:

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:56Z
Check the source to verify that this is the current version before use.

Material	Pattern nomenclature	Cut parts
Basic material	Top crown	1
	Top side	1
	Bottom side	1
	Side lining	1
	Top brim 1/	1
	Bottom brim 1/	1
Non-woven material	Brim interlining	1
	Headband stay	1

1/ Same pattern part.

3.5 Construction. The construction shall conform in all respects to the requirements specified in table I and herein. Figure I is furnished solely for guidance and information. Should variation from the written document appear in figure I, the document shall govern.

3.5.1 Stitches, seams, and stitching. All stitches, seams and stitching shall conform to FED-STD-751. The type of seam, stitching and stitches per inch shall be as specified in table I. Seam allowances shall be maintained with seams sewn so that no raw edges, run-offs, pleats, puckers, or open seams occur. When two or more methods of seams or stitches are given for the same operation, any one may be used. Where stitch type 401 is used, the looper (underthread) shall be on the inside of the hat.

3.5.1.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched not less than 1/4 inch except where ends are turned under or caught in other seams or stitching. Ends of a continuous line of stitching shall overlap not less than 1/2 inch. Thread tensions shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be imbedded in the materials sewed.

3.5.1.1.1 Repairs of type 301 stitching.

a. When thread breaks or bobbin run-outs occur during stitching, except prestitching, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch back of the end of the stitching. 1/

b. Except for prestitching, thread breaks, or two or more consecutive skipped or run-off stitches noted during inspection of the item shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch in back of the defective area, continue over the defective area, and continue a minimum of 1/2 inch

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:56Z
Check the source to verify that this is the current version before use.

beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.5.1.2 Type 101 and 401 stitching. Thread tension shall be maintained so that there will be no loose or excessively tight stitching resulting in puckering or the materials sewn. All repairs shall be in accordance with 3.5.1.1.1 a. and 3.5.1.1.1 b. Repairs to stitch type 401 may be accomplished by use of stitch type 301.

3.5.2 Bartacking. Bartacking shall be 1/4 ± 1/16 inch long, 1/8 ± 1/32 inch wide and shall contain 8-14 stitches. Bartacks shall be free from thread breaks and loose stitching.

3.5.3 Automatic stitching. Automatic machines may be used to perform any of the stitch patterns provided the requirements for the stitch pattern, stitches per inch, and size and type of thread are met; and at least three tying, overlapping or backstitches are used to secure the ends of the stitching.

3.5.4 Repairs. Repairs such as mends, darns, patches or splices are not permitted on the hats.

3.6 Manufacturing operations requirements. The hat shall be made in accordance with the operation requirements specified in table I. The contractor is not required to follow the exact sequence of operations listed, unless otherwise specified. Seam allowances required to be turned and seam turn-up of side and lining may be precreased. Any additional basting or holding stitches used to facilitate manufacture are permissible provided the thread does not show on the finished hat.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:56Z
Check the source to verify that this is the current version before use.

NO.	TABLE I. MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. a. Cut all parts in strict accordance with patterns furnished, which show size, directional lines (warp or filling direction) and notches for proper assembly of parts. b. Cut all parts of basic material from one piece of material except side lining and hem binding which may be cut from ends. The side lining, when cut from ends, shall approximate the shade of the respective basic material. c. Cut the brim binding on the bias, not less than 1-1/8 inches wide. The brim binding may be placed on rolls. d. Cut the lace for the chin strap with a heated blade to fuse ends of straps. The cut lengths shall finish 25 ± 1 inch.						
2.	Replacement of damaged parts. Care shall be exercised during spreading, cutting and manufacturing operations to assure that material defects and damages as specified in 4.2.3, are excluded and replaced with non-defective and properly matched material.						
3.	Marking. Mark, ticket or bundle all cut parts to insure a uniform shade, and size throughout the hat. Any method of marking may be used except:						

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:56Z
Check the source to verify that this is the current version before use.

TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
3.	Marking. (cont'd) a. Corrosive metal fastening devices. b. Sew-on tickets. c. Adhesive type tickets which leave an impression or traces of paper or adhesives on the material after removal of the ticket.						
4.	Make chin strap keeper and chin strap. a. Cut leather piece for chin strap keeper, 3/4 ± 1/8 inch long by 1/2 ± 1/8 inch wide. b. Punch two holes, equally speced, in keeper to accommodate chin strap. The holes shall be of sufficient length and width to insure proper function of the keeper, i.e., keeper shall not move without exertion of a slight pressure. c. Thread chin strap through holes in keeper.						
5.	Make brim. Finished appearance. The brim shall be made of two plies of basic material interlined with one ply of non-woven material and shall finish without twists or puckers. The brim shall be quilted with parallel lines of stitching and shall finish 2-3/8 ± 1/8 inches (see table II and finish with an upward curl). a. (1) Superimpose outside edge of bottom brim and brim interlining with notches matching, and stitch together 1/16 to 1/8 inch from outside edges.	101 or 301 or 401	SSa-1	6-14	50 50 50	 50 70	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 20260421T23:56Z
Check the source to verify that this is the current version before use.

NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
5.	Make brim. (cont'd)						
	(2) Superimpose outside edges of top brim and brim interlining with notches matching and stitch 1/16 to 1/8 inch from outside edges through all plies.	101 or 301 or 401	SSa-1	6-14	50 50 50	 50 70	
	or						
	b. Superimpose outside edges of bottom brim, brim interlining, and top brim with notches matching and stitch together 1/16 to 1/8 inch from outside edges in one operation.	101 or 301 or 401	SSa-1	6-14	50 50 50	 50 70	
	c. Stitch bottom brim to top brim along inner edge adjacent to brim interlining.	301		8-10	50	50	
	d. Single or double stitch brim plies together with the row of stitches starting from outside edge of brim and continue stitching for the entire brim width. The rows of stitching shall be spaced 3/16 inch to 1/4 inch apart. The needle thread shall finish on top brim in the finished hat.	301		8-10	50	50	
	e. Turn under one end of binding 1/4 inch and superimpose edges of binding on underside of brim and stitch binding to outside edge of brim 3/16 to 1/4 inch from edge, holding binding tight. The binding shall start and finish at center back (± 1/2 inch tolerance) with the turned end of binding lapped not less than 1/2 inch on the opposite end of binding.	301 or 401	BSg-2(a)	10-14	50 50	50 70	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:56Z
Check the source to verify that this is the current version before use.

NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
5.	Make brim. (cont'd)						
	f. Turn binding over edge of brim, turn in raw edge and stitch 1/16 to 3/32 inch from edge through binding and brim. Binding shall finish not more than 3/8 inch wide on top side of brim, and shall be so attached that brim will curve upward 3/4 to 1 inch.	301	BSg-2(b)	10-14	50	50	
	or						
	g. As an alternate the binding may be attached to the brim in one operation, using a folder.						
6.	Join sides.						
	a. Join top side to bottom side with front notches matching.	301 or 401	LSq-2(a)	10-14	50 50	50 70	
	b. Turn down bottom side and raise stitch bottom side, 1/16 to 3/32 inch from outside edge.	301	LSq-2(b)	10-14	50	50	
	c. Join center back seam of top and bottom sides. Open seam allowances and stitch through the turned edges 1/16 to 3/32 inch from each side of joining seam line.	301 or 401 and 301	SSa-1 SSz-3(b) shows finished seam	10-14 10-14	50 50	50 50	
	d. Stitch the instruction label on the inside of hat to the side lining on left side of hat, positioned 3/4 to 1 inch from back raw edge. Stitch label on all four sides.	301	LSbj-1	10-14	50	50	
	e. Join center back seam of side lining and open seam allowances.	301 or 401	SSa-1	10-14	50 50	50 70	

Source: https://assist.dla.mil -- Downloaded: 20261021T23:56Z
Check the source to verify that this is the current version before use.

TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
7.	Join sides to crown.						
	Finished appearance. The top side piece shall finish on top of crown and shall be free of twists, puckers, pleats or raw edges.						
	a. Join side lining to inside of crown, matching front notches and with back seam in line with back notch on crown. Stitching shall not show on outside.	101 or 301 or 401	SSa-1	10-14	50 50 50	 50 70	
	b. Join top side to outside of crown, matching front notches and with back seam in line with back notches on crown.	301 or 401	LSbp-2(a)	10-14	50 50	50 70	
	c. Turn down top side and raise stitch top side, 1/16 to 3/32 inch from edge.	301	LSbp-2(b)	10-14	50	50	
	NOTE: The stitching specified in operation b and c shall not show on inside of hat.						
8.	Join sides to brim.						
	a. Position ends of chin strap to underside of brim at each side notch and tack in position.	301 or 401 or bartack		6-14	50 50	50 70	
	b. Position bottom side piece to top of brim matching front notches and back seam in line with back notch of brim and raw edges even and stitch side piece to brim, not through brim interlining.	301 or 401	LSq-2(a)	10-14	50 50	50 70	
	c. Turn up bottom side piece and edge stitch side piece 1/16 to 3/32 inch from edge.	301 or 401	LSq-2(b)	10-14	50 50	50 70	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:56Z
Check the source to verify that this is the current version before use.

NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
8.	Join sides to brim. (cont'd)						
	d. Position headband stay piece to inside of side lining with bottom edge of stay piece 1/2 inch from bottom edge of lining and even with back seam of lining. Stitch top edge of stay piece to lining 1/16 to 1/8 inch from edge of stay piece.	301 or. 401	LSq-2(b)	10-14	50 50	40 70	
	e. Turn up bottom raw edge of lining over bottom edge of stay piece.						
	f. Position side lining to bottom of brim matching front notches and back seam in line with back notch of brim and stitch side lining to brim, 1/16 to 3/32 inch from turned edge, catching bottom edge of size label in the stitching. The label shall extend upward into the crown and positioned to right side of center back seam.	301		10-14	50	50	
	NOTE: The lining shall not be full or twisted on the finished hat nor shall it twist or cause fullness to the outside of the finished hat. The lining shall not extend below the brim.						
9.	Blocking.						
	Block crown on proper sized heated blocks and flange brims with metal dies on hydraulic press with heat and pressure to properly set brims.						

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:56Z
Check the source to verify that this is the current version before use.

NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
10.	Attach ventilators. Pre-punch holes through side and lining, as indicated by marks on pattern, and attach a ventilator at each hole with washer on inside of hat.						
11.	Clean hats. a. Trim all thread ends. b. Remove all loose threads, spots, stains, shade markings, and size markings.						

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:56Z
Check the source to verify that this is the current version before use.

3.7 Finished measurements. The finished hat shall conform to measurements for each respective size, as shown in table II.

TABLE II. Finished measurements (in inches)

Size	Inside hat (head measurements) 1/	Crown height at center front 2/	Crown height at center back 3/	Brim width 4/
6-3/8	20-1/2	2-3/4	3-1/2	2-3/8
6-1/2	20-7/8	all sizes	all sizes	all sizes
6-5/8	21-1/4			
6-3/4	21-5/8			
6-7/8	22			
7	22-3/8			
7-1/8	22-3/4			
7-1/4	23-1/8			
7-3/8	23-1/2			
7-1/2	23-7/8			
7-5/8	24-1/4			
7-3/4	24-5/8			
7-7/8	25			
Tolerance	± 1/4	± 1/4	± 1/4	±1/8

1/ Measure along bottom edge of side lining.

2/ From base of side to crown seam on outside of center front.

3/ From base of side to crown seam on outside of center back.

4/ From base of side to outer bound edge of brim.

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:56Z
Check the source to verify that this is the current version before use.

3.8 Workmanship. The finished hat shall conform to the quality of product established by this document. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.2.1.1 Certification. The ventilators shall be accepted on the basis of a contractor's certificate of compliance with the requirements specified in 3.2.8.

* 4.2.3 End item visual examination. The end item shall be examined for defects listed in table III. The lot size shall be expressed in units of hats. The sample unit shall be one hat. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects, 10.0 for major and minor A combined defects, and 15.0 for total (major, minor A, and minor B combined) defects.

TABLE III. End item visual defects

Defects	Classification Major	Minor A	Minor B
Blocking			
a. Omitted.	X		
b. Poorly blocked, causing distortion on crown or brim; hat not symmetrical, affecting appearance.			X

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:56Z
Check the source to verify that this is the currentversion before use.

TABLE III. End item visual defects (cont'd)

Defects	Classification Major	Minor A	Minor B
Cleanness			
a. Any spot or stain on outside.			X
b. Thread ends not trimmed, loose threads, or one or more shade or size marking not removed.			X
Component and assembly			
a. Any component part or required operation omitted (unless otherwise classified herein).	X		
b. Any operation not performed as specified (unless otherwise classified herein).			X
Cutting			
Any component part not cut in strict accordance with directional lines indicated on pattern or not cut in accordance with document requirements.	X		
Ventilators			
a. One or more omitted or added.	X		
b. Broken or bent.	X		
c. One or more improperly clinched, i.e., loose or raw edges exposed around edges of ventilator on outside.	X		
d. One or more not properly clinched, i.e., raw edges of lining exposed around edges of ventilator on inside.			X
e. Side puckered, pleated or full in clinching one or more ventilators.		X	
f. One or more ventilators with washer on outside of hat.	X		
Label (identification-size or instructions)			
a. Missing, incorrect, or illegible.	X		
b. Not securely attached.		X	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:56Z
Check the source to verify that this is the current version before use.

TABLE III. End item visual defects (cont'd)

Defects	Classification Major	Minor A	Minor B
Material defects and workmanship damages			
a. Any hole or shade bar.	X		
b. Any defect such as a smash, multiple float or slub.	X		
c. Cut, tear, tip, mend or burn.	X		
d. Needle chew.	X		
Seams and stitching			
a. General:			
1. Seam or component part twisted, puckered or pleated.		X	
2. Part of material caught in any unrelated operation or stitching.	X		
3. Ends of stitching not secured as specified.		X	
4. Thread break (all stitch types) stitched less than 1/2 inch beyond each end of break.			X
b. Gage of stitching and seam allowances:			
1. Irregular, i.e., unevenly gaged, or corresponding stitchings not uniformly gaged (to be scored only when condition exists along major portion of seam).		X	
2. Width not as specified or not within range of gage specified or varies more than 1/16 inch when no range is specified.			X
NOTE: The above defects to be scored only when the condition exists on major portion (i.e., over one half or more) of seam.			
c. Open seams (one or more broken or two or more consecutive skipped or runoff stitches constitute an open seam):			
1. Any seam open on outside:			
- up to 1/4 inch inclusive			X
- more than 1/4 inch	X		

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:56Z
Check the source to verify that this is the current version before use.

TABLE III. End item visual defects (cont'd)

Defects	Classification Major	Minor A	Minor B
Seams and stitching (cont'd)			
2. Any seam open on inside of hat:			
- up to 1/4 inch inclusive			X
- more than 1/4 inch to 1 inch inclusive		X	
- more than 1 inch	X		
d. Raw edges:			
1. Any raw edge on outside of hat:			
- 1/4 to 1/2 inch inclusive		X	
- more than 1/2 inch	X		
2. Any raw edge on inside of hat more than 1/4 inch.			X
NOTE: Raw edges not securely caught in stitching shall be classified as an open seam.			
e. Runoffs:			
1. On joining seams, when resulting in an open seam, score as "open seam".			
2. On binding stitching:			
- 1/4 to 1/2 inch inclusive			X
- over 1/2 inch		X	
f. Seam and stitch types:			
1. Not specified seam or stitch type.		X	
2. Looper thread on outside (when 401 stitch is used).	X		
3. Chain of 101 stitch type exposed on outside of hat.	X		
g. Stitch tension:			
1. Loose tension resulting in a loose seam.		X	
2. Loose tension on top quilting stitches or edge stitching, resulting in loosely exposed lower thread on top thread.		X	
3. Tight tension (stitches break when strain is applied to the direction of seam).	X		

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:56Z
Check the source to verify that this is the current version before use.

TABLE III. End item visual defects (cont'd)

Defects	Classification Major	Minor A	Minor B
Seams and stitching (cont'd)			
h. Stitches per inch (to be scored only when the condition exists on major portion of seam or stitching):			
1. Less than minimum specified.		X	
2. More than maximum specified.			X
Shaded parts			
a. Badly shaded parts (except those cut from ends).	X		
b. Parts permitted to be cut from ends badly shaded.			X
Brim			
a. Twisted, stretched or poorly shaped.	X		
b. Rows of quilting stitches less than 3/16 inch or more than 1/4 inch apart.		X	
c. Needle thread of quilting stitching finished on underside of brim.			X
d. Fullness between rows of quilting stitches.		X	
e. Binding, twisted, puckered or pleated.	X		
f. Outside end of binding not turned under or not within 1/2 inch of center back of brim.			X
g. Binding finished more than 3/8 inch wide on top of brim for a distance of 2 inches or more.		X	
h. Irregular in width by more than 1/2 inch inclusive.		X	
i. Ends of binding lapped less than 1/2 inch.		X	
j. Brim curving upward less than 3/4 inch or more than 1 inch.		X	
Crown and sides			
a. Crown twisted or distorted.		X	
b. Side twisted or distorted.	X		
c. Side lining twisted or distorted.			X
d. Lining extending below edge of brim.			X
e. Side back seam and side lining back seam, not seam on seam, by more than 1/8 inch.		X	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:56Z
Check the source to verify that this is the current version before use.

TABLE III. End item visual defects (cont'd)

Defects	Classification Major	Minor A	Minor B
Crown and sides (cont'd)			
f. Stitches joining top side to crown exposed on inside of hat.		X	
g. Stitches joining side lining to crown exposed on outside of hat.		X	
h. Joining seams of top side to bottom side out of alignment by more than 1/16 inch at back seam.		X	
Chin strap and keeper			
a. Chin strap missing.	X		
b. End of chin strap not fused.		X	
c. Keeper missing.	X		
d. Keeper less than 5/8 inch or more than 7/8 inch long; or less than 3/8 or more than 5/8 inch wide.			X
e. Holes in keeper too small or too large not permitting chin strap to function properly, i.e., falls freely from set position or keeper immovable.	X		
f. Chin strap measuring less than 24 inches or more than 26 inches long.			X
g. Chin strap twisted.			X

4.2.3.1 End item dimensional examination. The end item shall be examined for conformance to table II. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of hats. The sample unit shall be one hat. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.2.4 Packaging inspection. An examination shall be made to determine that the preservation, packing and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged with the exception that it need not be closed. Examination for closure defects listed below shall be made on shipping containers fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:56Z
Check the source to verify that this is the current version before use.

Examine	Defect
Marking (exterior interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Material	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flap, loose strapping, inadequate stapling, or improper taping. Bulged or distorted container. Open and noncontinuous heat-sealed seams and closure of polyethylene bag. Incorrectly fabricated bag. Alternative closure of polyethylene bag not provided with a twist tie.
Content	Number of intermediate packs per shipping container is more or less than required. Number of hats per intermediate pack is more or less than required. 1/

1/ For this defect, one intermediate pack from each container in the sample shall be examined.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. Each hat shall be firmly rolled along the lengthwise dimension of the hat. Each hat shall be inserted into a snug-fitting polyethylene film bag of 0.00125 inch thickness (± 25 percent tolerance). The bag may be fabricated from polyethylene film tubing or sheeting. The bag shall be formed with heat sealed seams that are straight, continuous and parallel to each other and the formed edges of the bag. The final closure of the bag shall be heat sealed with the seal made as close as possible to the open end. Alternatively, bag closure may be accomplished by a twist tie method. Prior to or during the closure operation, excess air within the bag shall be expelled.

5.1.1.1 Intermediate pack. Twelve hats of one size only, unit packed as specified in 5.1.1, shall be placed in an intermediate fiberboard box conforming to style FOL (end-opening), grade W6c or W6s of PPP-B-636. The hats shall be placed flat one in length, one in width and twelve in depth. Inside dimensions of the

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:56Z
Check the source to verify that this is the currentversion before use.

fiberboard box shall be 14-1/2 inches in length, 4 inches in width and 13 inches in depth. Each fiberboard box shall be closed in accordance with method IV as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.4.

5.1.2 Commercial. Hats shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A packing. One hundred and forty-four hats of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Level A intermediate packs shall be packed three in length, four in width, and one in depth. Inside dimensions of each container shall approximate 44 inches in length, 16-1/2 inches in width and 14 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.4. Toward the end of the contract, or when there are less than the required amount per container of the same size, mixed sizes may be packed in the same shipping container.

5.2.2 Level B packing. One hundred and forty-four hats of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Level A intermediate packs shall be packed three in length, four in width, and one in depth. Inside dimensions of each container shall approximate 44 inches in length, 16-1/2 inches in width and 14 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.4. Toward the end of the contract, or when there are less than the required amount per container of the same size, mixed sizes may be packed in the same shipping container.

5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.4.

5.2.3 Commercial packing. Hats preserved as specified in 5.1 shall be packed in accordance with MIL-STD-1188.

5.3 Marking. In addition to any special marking required by the contract or purchase order, intermediate packs and shipping containers shall be marked in accordance with MIL-STD-129 and MIL-STD-1188 as applicable.

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:56Z
Check the source to verify that this is the current version before use.

5.3.1 Labels, mixed sizes. Each shipping container packed with mixed sizes shall have securely attached to the end and side, directly under the printing or stenciling, a white paper label 5- by 4-inches with the words "MIXED NSN's" plainly stamped or printed thereon and under these words shall be legibly stamped or printed the correct quantity and NSN's contained therein.

5.3.2 Polyethylene bagged unit packs. Polyethylene bagged unit packs shall have the required identification information legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification.

6. NOTES

6.1 Intended use. The hat is intended for use by military personnel of the Department of Defense for special purposes in desert environments.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this document.
(b) Size required (see 1.2).
(c) Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
(d) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.3 Samples. For access to sample of the hat and shade samples of the materials, address the contracting activity issuing the invitation for bids.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.2).

6.5 Supersession data. This document includes the requirements of Limited Production Purchase Description 40-71, Hat, Camouflage Pattern: Desert.

Custodians:

Army - GL
Navy - MC
Air Force - 99

Preparing activity:

Army - GL

Project No. 8415-0340

Review activities:

Army - MD
Navy - NU
Air Force - 82
DLA - CT

User activity:

Air Force - 45

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:56Z
Check the source to verify that this is the currentversion before use.

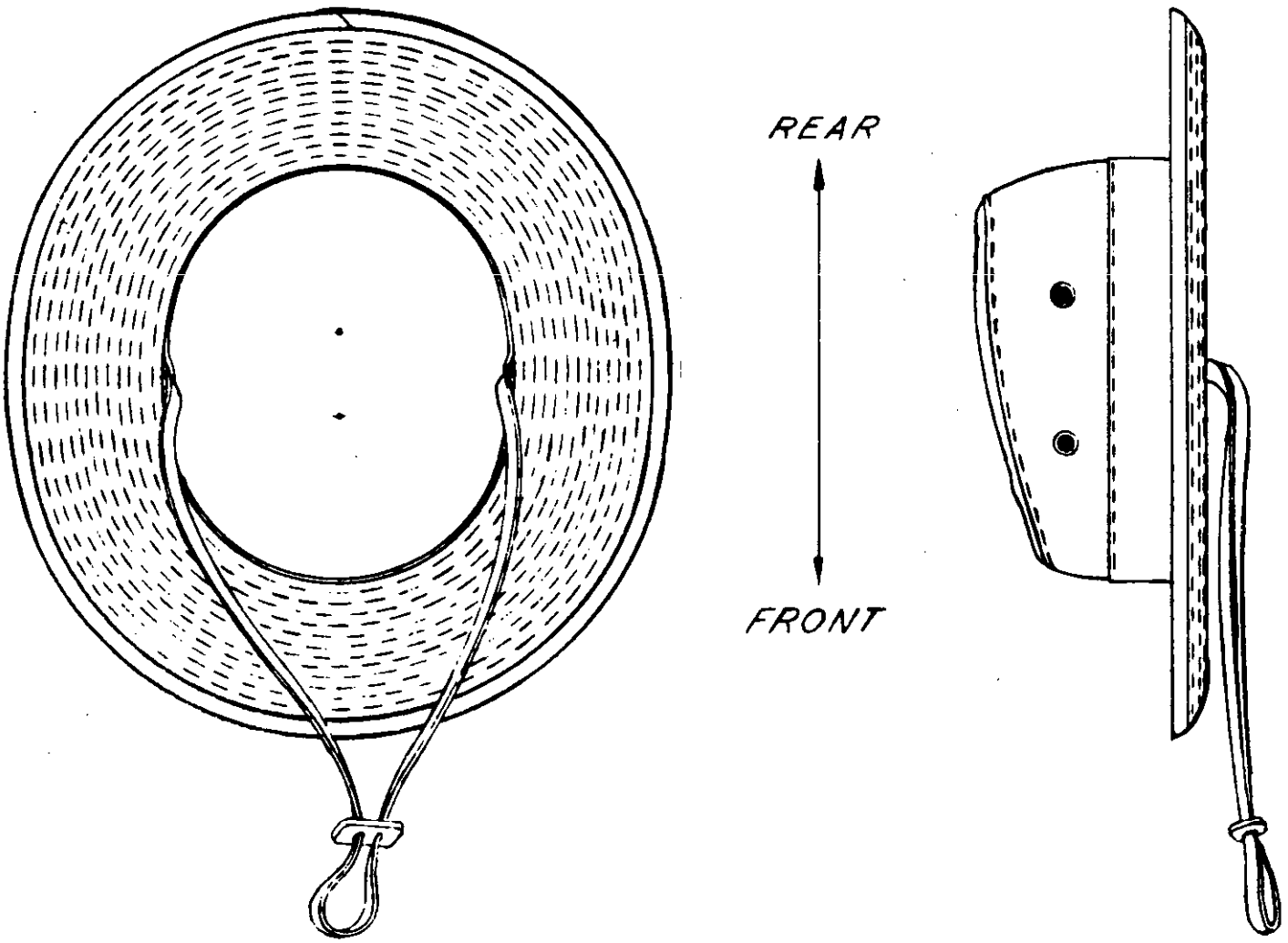


FIGURE 1. HAT, CAMOUFLAGE PATTERN: DESERT

Source: https://assist.dla.mil -- Downloaded: 20261O-21T23:56Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-44105	Hat, Camouflage Pattern: Desert

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

NATICK OP, 1 APR 83

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:56Z
Check the source to verify that this is the current version before use.